

Battery Data Analysis

*Insights from Swap Data and Battery
Features Data*

Introduction

Purpose: To analyze battery data, identify low-usage batteries, and forecast future usage time.

Objective:

- To identify batteries with low usage time
- Use battery data to predict future usage patterns
- Apply time series analysis to forecast battery life

Data Overview

- **Data Sources:**
 - **Swap Data:** Contains records of battery swaps
 - **Battery Features Data:** Includes attributes like capacity, cycles, and performance metrics
- **Key Variables:**
 - **Swap in & swap out time**
 - **Battery Features:** E.g. capacity, mileage, SOC, temperature during operation, etc

Methodology

1: Identifying Low-Usage Batteries

- Analyzed swap data to determine average usage time between swaps
- Grouped data by battery ID and month
- Flagged batteries with usage times below a certain statistical threshold
- Flagged batteries with a fast-declining usage time

2: Feature Integration : Merged swap data with battery feature data to understand factors influencing usage time.

3: Forecasting Usage Time : Applied regression models to predict future usage time based on features.

4: Time Series Analysis : Used ARIMA to model and forecast battery life trends.

Exploratory Data Analysis

❖ Insights from Swap Data:

- Plotted the distribution of battery usage times using histogram.
- Identified 22 batteries with low usage times (with z-scores less than -2)
- Identified 75 batteries with fast-declining average usage time.

❖ Insights from Feature Analysis:

Key drivers of low usage:

- Low total charged capacity
- Alarm flag - these batteries had low values of "AlarmDesc" and "AlarmFlag" - we shall further explore the meaning of this.
- Low maximum charge current

Model Building and Evaluation

Regression Models:

- ❖ Used regression for predicting the expected average usage time between swaps
- ❖ **Features Used:**
 - Board surface temperature
 - Cell series number
 - Internal temperature of battery
 - Mileage
 - Number of cycles
 - Battery SOC
 - Average speed information
 - Total current
 - Total discharged capacity
 - Total voltage
 - Alarm flag
 - Battery capacity
 - Charge protect voltage
 - **Evaluation:** Achieved a mean squared error (MSE) of 1.35 on test data

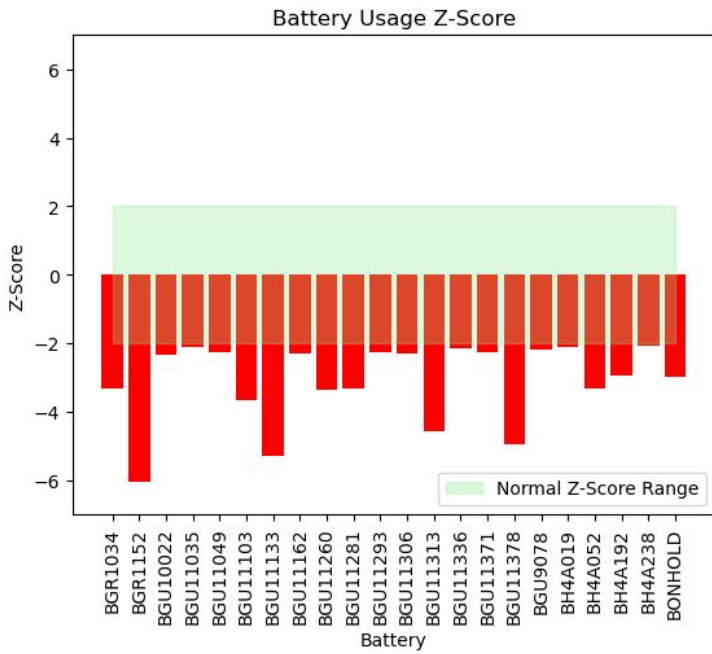
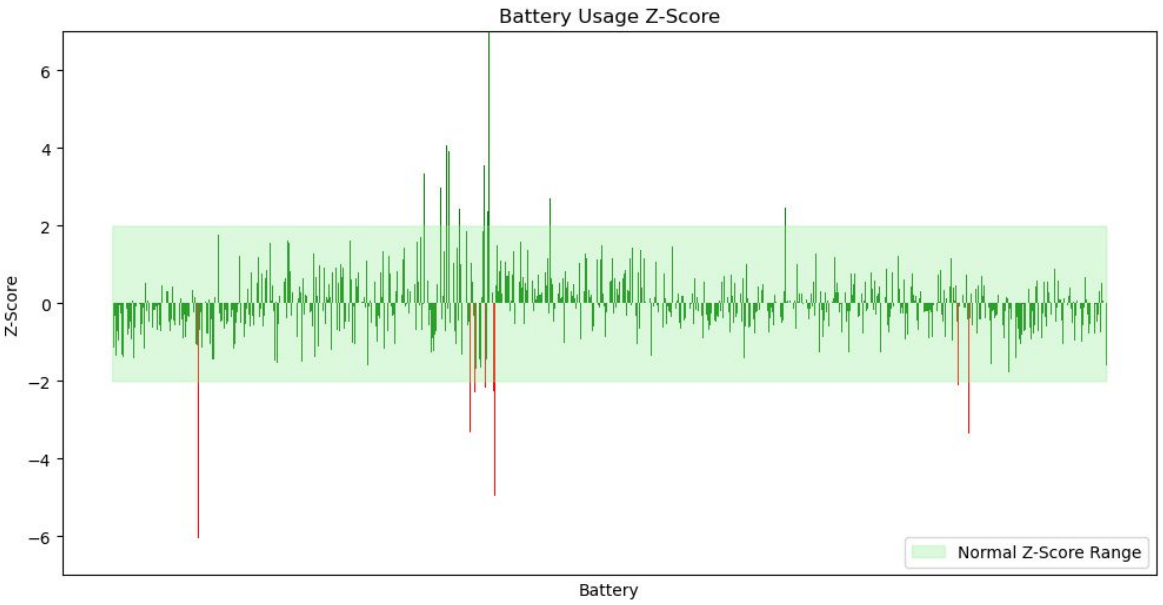
Time Series Models:

- ❖ Applied ARIMA to predict term battery life trends
- ❖ **Evaluation:** Achieved a MAE of 4.34

Results

1. Identified Low-Usage Batteries:

- The graph on the left shows the z-scores of all batteries. The red spikes show the specific batteries with extremely low z-score, meaning they have a low average usage time.
- **The green shaded area shows the normal z-score range.**
- The graph on the right-hand side shows batteries with significantly low average usage time between swaps.



2. Time Series Forecast: Forecasted battery average usage hours for the next 10 months, stored as a csv file

Conclusion

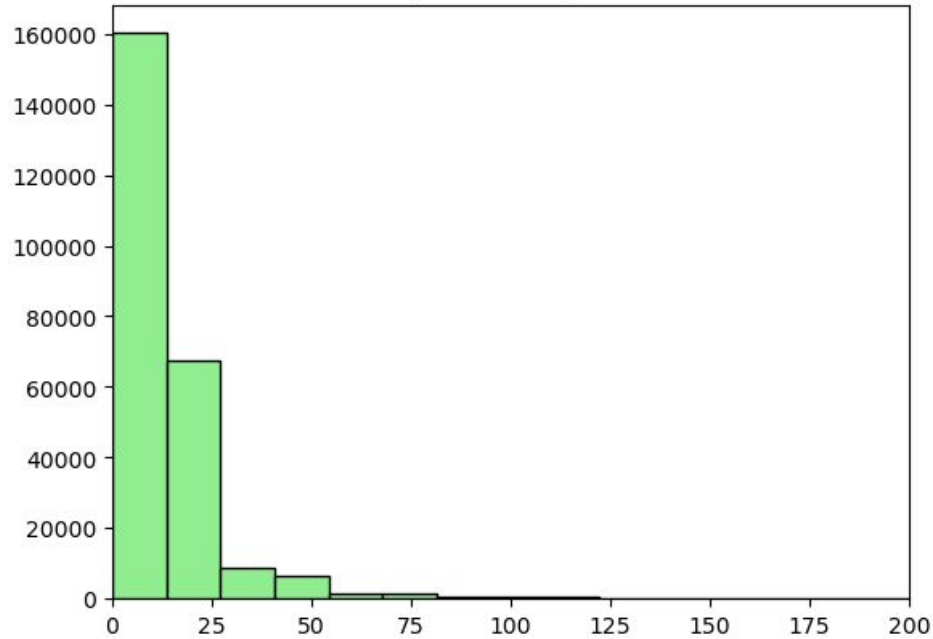
- **Key Findings:**
 - 22 (1.36)% of batteries identified as low usage
 - 75 (4.64%) of batteries identified as fast-declining
 - Key factors affecting usage time: charged capacity, alarm flag, and maximum charge current
 - The time series forecast provides insights into what would be expected of battery performance trends in the near future
- **Recommendations:**
 - **Charged capacity** - seen as these batteries have a low charged capacity, it is possible there has cases of overcharging which affects battery performance. The battery maintenance team should ensure that batteries are not overcharged, given their specific features.
 - **Alarm Flags** - in addition to the above, the team should pay closer attention to alarm flags and resolve any issues as soon as they arise.
 - **Maximum charge current** - To ensure the maximum charge current is not exceeded, the team should ensure the charging system used for each battery has a maximum current rating that matches or is lower than the battery's recommended charge current.
- **Next Steps / Model Improvement:**
 - The models reliability can/should be improved by using more granular data
 - Perform feature importance analysis and feature selection to improve model performance
 - Extend the time series model to predict long-term battery life using more granular data

The End

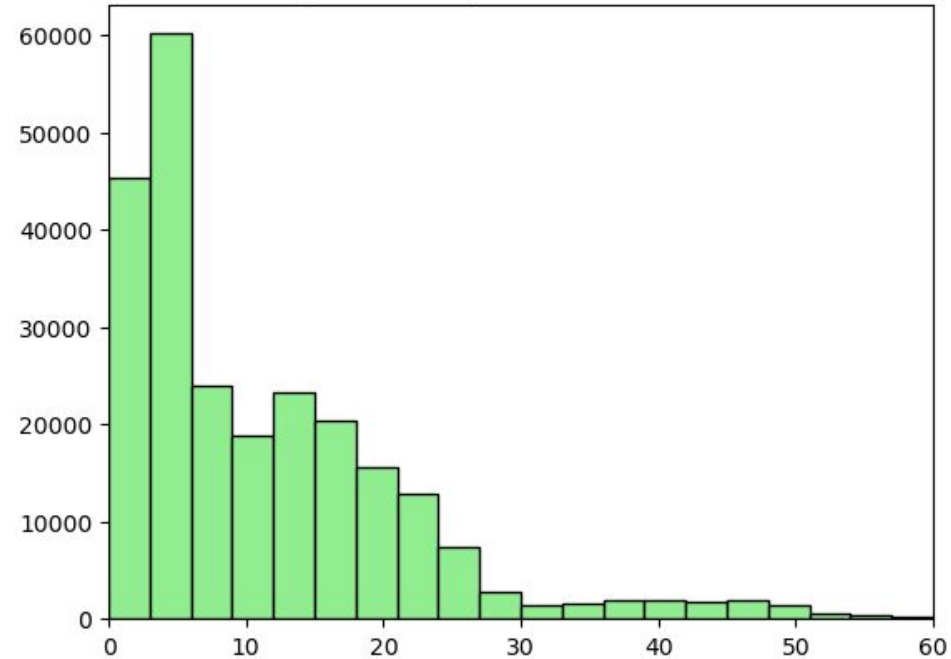
Thank you!

Appendix

Histogram of usage hours



Histogram of usage hours without outliers



Distribution of battery usage time

- Most batteries last between 0 and 30 hours before they need a recharge.
- Only a few go over 30 hours, with extreme cases taking over 50 hours
- Note: these extreme cases could mean the battery is idle.